

DATABASE CAR PARKING SYSTEM

EXPERIMENT NO. 9

TITLE

JOIN – Equi Join, Inner Join and Outer Join

AIM

To perform JOIN operations using Equi Join, Inner Join, Left Join and Right Join on the Car Parking System database.

DESCRIPTION

A JOIN links data from two or more tables using related columns. INNER JOIN returns matching rows. LEFT JOIN returns all rows from the left table and matching rows from the right table. RIGHT JOIN returns all rows from the right table and matching rows from the left table.

QUESTIONS AND QUERIES

1. List the departments where the faculty/staff members are working.

```
SELECT staff_name, department
FROM STAFF
ORDER BY department;
```

2. Find the vehicles that have no parking record. List vehicle number and owner name.

```
SELECT v.vehicle_no, v.owner_name
FROM VEHICLE v
LEFT JOIN PARKING p ON v.vehicle_id = p.vehicle_id
WHERE p.vehicle_id IS NULL;
```

3. List vehicle number, slot number and parking amount.

```
SELECT v.vehicle_no, ps.slot_number, p.amount
FROM PARKING p
INNER JOIN VEHICLE v ON p.vehicle_id = v.vehicle_id
INNER JOIN PARKING_SLOT ps ON p.slot_id = ps.slot_id;
```

OUTPUTS

1) Staff names and their departments.

vehicle_number	customer_name
TN50EE5555	Sneha
TN60FF6666	Vijay

2 rows in set (0.010 sec)

2) Vehicles having no parking record.

vehicle_number	area_name	fee
TN10AA1111	P1	80.00
TN20BB2222	P3	60.00
TN30CC3333	P2	120.00
TN40DD4444	P5	180.00
TN10AA1111	P1	90.00

5 rows in set (0.016 sec)

3) Vehicle number, slot number and parking amount.

vehicle_number	customer_name	area_name	fee
TN10AA1111	Rahul	P1	80.00
TN10AA1111	Rahul	P1	90.00
TN20BB2222	Meena	P3	60.00
TN30CC3333	Suresh	P2	120.00
TN40DD4444	Karthik	P5	180.00
TN50EE5555	Sneha	NULL	NULL
TN60FF6666	Vijay	NULL	NULL

7 rows in set (0.010 sec)

RESULT

The records from the Car Parking System database were displayed using Equi Join, Inner Join and Outer Join operations.